gastro intestinal tract where most of the digestion and absorption occurs. The small intestine has a large surface area due to the presence of villi or microvilli to aid absorption of nutrients.

The small intestine is divided into three regions: The duodenum, a C shaped tube, which starts at the pyloric sphincter that extends about 25 cm (10 inches) and merges with the jejunum. The jejunum is about 1 m (3 ft) long and extends to the ileum, the final and longest region of the small intestine, measuring about 2 m (6 ft); ileum joins the large intestine at a smooth muscle sphincter called the ileocecal sphincter (valve) (Figure 2.4).

Structure of Villi

An intestinal villus is a small finger like projection that extends into the lumen of the small intestine (Figure 2.5). Each villus has many microvilli projecting from its epithelial surface, collectively forming a brush border. The structure of villi aids in absorption of nutrients by providing large surface area and also have thin wall that reduces the distance for the nutrients to move across the intestine by a process called diffusion. Between the villi there are crypts, called crypts of Lieberkuhn, which are short glands. The Gut associated lymphoid tissue or the Peyer's patches are present in the small intestine and protect the intestine from the foreign pathogens.

Each villus consists of one cell thick thin epithelium surrounding the blood capillary and tiny lymphatic vessels called lacteals.

2.1.1.6 The Large Intestine

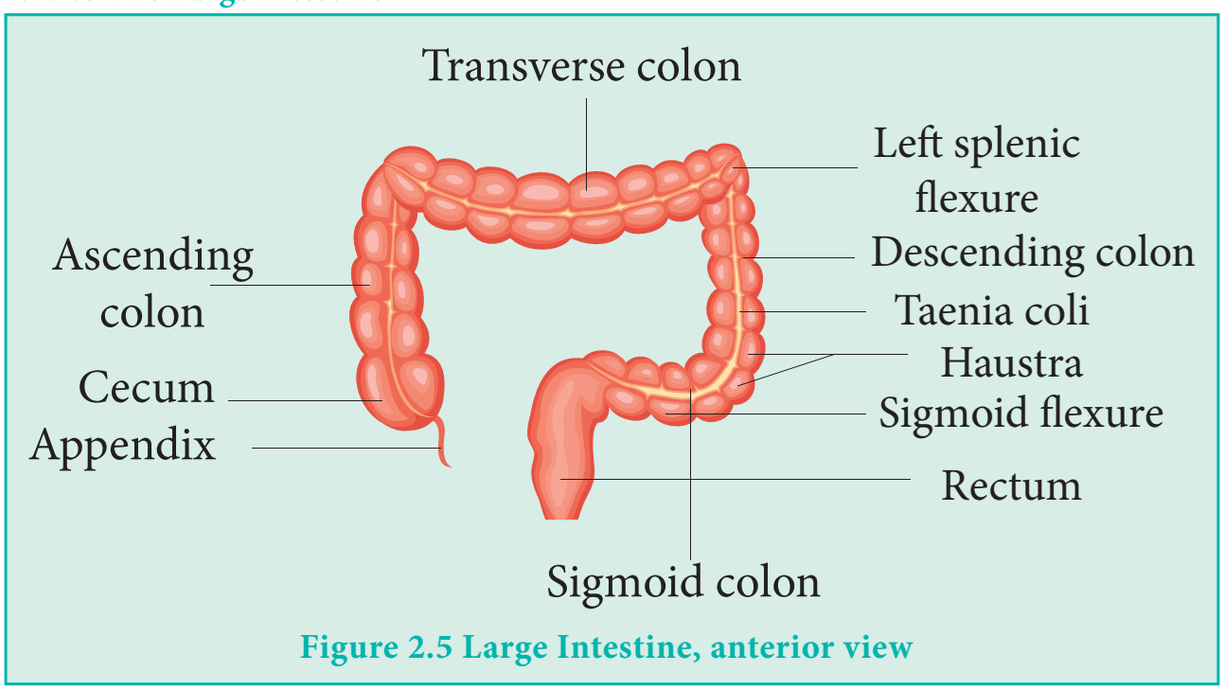


Figure 2.5 Large Intestine, anterior view

The large intestine includes the cecum, colon, rectum and the anal canal (Figure 2.5). It is larger in diameter than the small intestine (6.5 cm compared to 2.5 cm), but it is shorter in length. The large intestine absorbs water, salts, and some vitamins. The indigestible material is stored in the large intestine till it is eliminated via the anus. The cecum has a small projection called the vermiform appendix (vermiform -wormlike). The

XII U2 Digestion.indd 30 4/8/2020 3:05:22 PM